

Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence

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Abstract

This paper examines changes in the labor market for occupations exposed to generative artificial intelligence using high-frequency administrative data from the largest payroll software provider in the United States. We present six facts that characterize these shifts. We find that since the widespread adoption of generative AI, early-career workers (ages 22-25) in the most AI-exposed occupations have experienced a 13 percent relative decline in employment even after controlling for firm-level shocks. In contrast, employment for workers in less exposed fields and more experienced workers in the same occupations has remained stable or continued to grow. We also find that adjustments occur primarily through employment rather than compensation. Furthermore, employment declines are concentrated in occupations where AI is more likely to *automate*, rather than *augment*, human labor. Our results are robust to alternative explanations, such as excluding technology-related firms and excluding occupations amenable to remote work. These six facts provide early, large-scale evidence consistent with the hypothesis that the AI revolution is beginning to have a significant and disproportionate impact on entry-level workers in the American labor market.

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firm regardless of AI exposure. For workers aged 22-25, we find a 12 log-point decline in relative employment for the most AI-exposed quintiles compared to the least exposed quintile, a large and statistically significant effect. Estimates for other age groups are much smaller in magnitude and not statistically significant. These findings imply that the employment trends we observe are not driven by differential shocks to firms that employ a disproportionate share of AI-exposed young workers.

Fifth, the labor market adjustments are visible in employment more than compensation. In contrast to our findings for employment, we find little difference in annual salary trends by age or exposure quintile, suggesting possible wage stickiness. If so, AI may have larger effects on employment than on wages, at least initially.

Sixth, the above facts are largely consistent across various alternative sample constructions. We find that our results are not driven solely by computer occupations or by occupations susceptible to remote work and outsourcing. We also find that the AI exposure taxonomy did not meaningfully predict employment outcomes for young workers further back in time, before the widespread use of LLMs, including during the unemployment spike driven by the COVID-19 pandemic. The patterns we observe in the data appear most acutely starting in late 2022, around the time of rapid proliferation of generative AI tools.⁴ They also hold for both occupations with a high share of college graduates and ones with a low college share, suggesting deteriorating education outcomes during COVID-19 do not drive our results. For non-college workers, we find evidence that experience may serve as less of a buffer to labor market disruption, as low college share occupations exhibit divergent employment outcomes by AI exposure up to age 40.

While we caution that the facts we document may in part be influenced by factors other than generative AI, our results are consistent with the hypothesis that generative AI has begun to affect entry-level employment. We intend to continue to track the data on an ongoing basis to assess whether these trends change in the future.

Why might AI adversely affect exposed entry-level workers more than other age groups? One possibility is that, by nature of the model training process, AI replaces codified knowledge, the “book-learning” that forms the core of formal education. AI may be less capable of replacing

⁴OpenAI introduced ChatGPT in November 2022.

tacit knowledge, the idiosyncratic tips and tricks that accumulate with experience.⁵ As young workers supply relatively more codified knowledge than tacit knowledge, they may face greater task replacement from AI in exposed occupations, leading to greater employment reallocation (Acemoglu and Autor, 2011). In contrast older workers with accumulated tacit knowledge may face less task replacement. These benefits of tacit knowledge may accrue less to non-college workers in occupations with low returns to experience. Furthermore, more experienced workers may be more skilled in other ways, making them less vulnerable to substitution by AI tools (Ide, 2025). An important direction for research is to further model and test these predictions.

2 Related Literature

This paper engages with a large public debate in academia, public policy, business, and media on the employment effects of artificial intelligence. Much of this discourse centers on whether AI is displacing workers in exposed professions such as software engineers.⁶ Some work has noted that the unemployment rate for college graduates has risen above the rate for non-graduates, suggesting this as evidence of employment disruptions from AI (Thompson, 2025). Others have noted that these trends long preceded the spread of AI and have noted that publicly available data such as the Current Population Survey (CPS) show mixed evidence on employment changes in AI-exposed occupations (Lim et al., 2025; The Economist, 2025; Smith, 2025; Eckhardt and Goldschlag, 2025; Frick, 2025).⁷ These debates remain unsettled and in search of high quality data on labor market changes among exposed groups. Our paper provides large-scale data to measure employment changes with a high degree of granularity and precision, finding that young workers in AI-exposed occupations have indeed experienced employment declines.

⁵Ironically, one of the practical skills more likely to be learned on the job than in university computer science classes may be how to use AI software development.

⁶Some recent media on this topic includes Horowitch (2025); Ettenheim (2025); Raman (2025); Roose (2025); Peck (2025); Hoover (2025); Milmo and Almeida (2025); Wu (2025); Raval (2025). A number of technology executives have also warned of potential job loss from AI (Allen, 2025; Sherman, 2025; Bacon, 2025) or laid off workers with the aim of increasing AI investments (Jamali, 2025).

⁷Reports from industry have also shown mixed findings. Job posting platform TrueUp suggests a recent increase in postings in the tech sector (Lenny Rachitsky, 2025). On the other hand, Revelio Labs finds a decline in job postings, with the decrease steeper for entry-level workers (Simon, 2025). Indeed job posting data suggest declines in postings for new graduates but find these declines for less AI-exposed occupations as well (Lim et al., 2025). Chandar (2025b) notes that the correlation between job postings and employment has been weak over recent years. SignalFire finds steep declines in new graduate hires in the tech sector compared to pre-Pandemic levels (Doshay and Bantock, 2025), consistent with the findings in this paper. Data from Gusto also suggests a decline in new graduate hiring (Bowen, 2025).